

Initial Plan Report — Project Xceed

Seat Belt Detection System with Safety Analytics Integration

To: Designated MSIL Project Mentor
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1. Problem Understanding

This project aims at designing a camera system capable of distinguishing in real-time if all passengers in the car – from the driver's seat to the third row – have correctly fastened their seat belts. Any unfastened passenger must be immediately informed. The whole process should be implemented using only edge devices, without access to the Internet.

This may seem simple enough on the surface; however, it actually presents quite a number of serious problems:

Key Challenges

- **Speed on limited hardware:** The algorithm must work on relatively inexpensive edge equipment and operate with speed not less than 20 FPS. That is very important for any architecture decisions because it limits computational resources for this problem.
- **Extreme lighting conditions:** There might be a large variation between the brightest sunlight up to 25,000 lux and total darkness in the car cabin at night with less than 1 lux. This range is much larger than what off-the-shelf systems provide - they usually support either one or another condition but never both.
- **Visual ambiguity:** Sometimes it is quite difficult to tell apart a seat belt from something else, such as dark bags strapped to shoulders, dark clothing, and lanyards. Therefore, this model should understand when people have fastened their seat belt in the correct way.
- **Passengers clipping seat belts:** As mentioned before, some passengers simply clip seat belts behind them to silence alerts. Therefore, simple detection that a passenger fastened their seat belt is not sufficient for this problem.
- **Data pipeline:** In addition to seat-belt violation detection, I would like to collect this data along with timestamp and seat positions, and visualize them on a dashboard.

2. My Approach - Full-Stack AI Integration

Rather than building just a detection module, I'm designing the system in four layers that work together end-to-end:

- **AI Detection:** YOLOv8n will be used as the base model. This particular model is fast, compact, and ideal for running on the edge. To meet the required frames per second, I will optimize it using ONNX Runtime. The model will be trained to identify whether the child seat is "properly worn," "clipped behind," or is a "decoy object."
- **Hardware Alerting:** Upon detecting any violation, the AI engine triggers a GPIO/UART signal to the Arduino or LPC1768 board, which turns on the buzzer and LED lights right away. Placing everything in a microcontroller ensures zero latency since there will be no Operating System between detection and alerting.

- **Backend & Logging:** A minimalistic backend will be implemented using FastAPI, which will run on the local Pi machine and log every violation found into a SQLite database containing details such as seat positions, timestamps, and types of violations recorded.
- **Dashboard:** Finally, a React front-end that communicates with the local API endpoint and displays real-time information about the passengers' compliance, past logs, and overall system status.

3. Hardware I'm Requesting (With Justification)

Here is what I am going to ask and my reasoning for each component, with the goal of meeting all project requirements while staying within the ₹10,000 budget limit:

Component	Why I Need It
Raspberry Pi 5 (4GB)	It is the only affordable edge device that will be able to achieve ≥ 20 FPS performance with deep learning running on the CPU and simultaneously run a local backend and dashboard.
RPi Camera Module 3 NoIR	It allows achieving the minimum < 1 lux performance needed in the scenario where the cabin is lit entirely by nightlight.
Infrared (IR) LED Array	Since the RPi camera does not have an IR filter, the IR LED array serves as the primary source of lighting at such low lux values. The emitted light is not visible to human eyes.
LPC1768 / Arduino UNO	To ensure immediate triggering of the buzzers regardless of the operating system performance, it makes sense to put it on a separate microcontroller.
27W USB-C Power Supply	At the maximum level of computation, it is important to maintain a stable FPS rate, so providing more energy than the default power supply is crucial.
Buzzer & LED Kit	Hardware components that will provide visual and audio feedback once the passenger fails to wear their seatbelt properly.

4. Week-by-Week Plan

The final review will take place on June 2nd. Here is how I intend to organize the work:

- **Week 1 (Apr 6 - Apr 12):** Prepare the plan, purchase hardware, set up the environment, and initialize FastAPI + React framework locally.
- **Weeks 2 & 3 (Apr 13 - Apr 26):** Gather, label, and train YOLOv8n neural network using custom and Kaggle/Roboflow datasets for the three classes of images.
- **Week 4 (Apr 27 - May 3):** Improve the model by converting it into ONNX format and integrate into backend code. Submit the Mid Progress Report.
- **Weeks 5 & 6 (May 4 - May 17):** Transfer all files to Raspberry Pi 5 and perform the benchmark. Also test in dark conditions with RPi NoIR camera and IR LEDs.
- **Weeks 7 & 8 (May 18 - May 31):** Test under various lighting conditions ranging from full daylight to minimum < 1 lux. Identify any potential failures and improve dashboard UI.
- **Week 9 (Jun 1 - Jun 2):** Final test, make a video demonstrating the project's work, and complete the Technical Final Report.

5. What I'm Committing to Deliver

By the final review, the project should have:

- **Prototype:** Actual Edge device with real-time detection connected to an actual buzzer for non-compliance.
- **Master script:** Only one master script needed to start detection, record database logs, and display the dashboard - no time wasted setting up during demonstration.
- **Full documentation:** All the information needed to understand how the system works and recreate it yourself, including diagrams and API documentation.
- **Verified Performance:** At least 20 FPS performance verified for the Raspberry Pi 5, tested throughout all values of the specified lux range.